

Transient analyses of an interceptor trench

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Abstract. Unlike radial flow to a well the pattern of flow to a trench changes with time. Flow vectors at early times are normal to the surface of the trench and affect a small area, whereas after a long period of pumping they converge radially from large distances. This changing flow pattern controls performance, and ignoring the transient effects can result in errors when predicting flows in the vicinity of a trench. The method of instantaneous source functions is used to obtain transient analytical solutions to a variety of two- or three-dimensional problems related to interceptor trenches. Drawdown in the vicinity of a trench that is pumped at constant rate and discharge from a trench that is held at constant drawdown are given as functions of time. Travel times to trenches in regional flow fields are also obtained and expressed in dimensionless form. Applications of the results include improved predictions of the performance of environmental applications of interceptor trenches and estimation of aquifer parameters from the analysis of pump tests from trenches.

Introduction

Gravel-filled trenches are widely used to form highly permeable curtains that intercept contaminated groundwater or enhance the recovery from sites underlain by tight formations [Day, 1991; Meiri *et al.*, 1990; Mildenerger and Moran, 1987; Kufs *et al.*, 1983], but many of the methods available to predict their performance were published years ago to analyze dewatering or irrigation projects (a summary of published solutions is given by Beljin and Murdoch [1993]). Analyses conducted for dewatering or irrigation are typically derived to predict the location of a piezometric surface, so that predictions of travel paths and arrival times of contaminants are unavailable or limited to a few cases [e.g., Zheng *et al.*, 1988].

Many of the commonly cited analyses, including some recent studies, treat a trench in section view by assuming that it is infinitely long [Chambers and Bahr, 1992]. This assumption is valid early in the pumping history, and it may provide useful results for problems particularly suited to analysis in section view, such as the treatment of dividing streamlines beneath a trench [Zheng *et al.*, 1988]. However, the assumption of infinite length results in errors that increase as pumping continues and water is recovered beyond the ends of the trench, and it may cause significant errors as the system approaches steady state. These errors are illustrated by considering the steady state discharge Q_{s1} from a trench of length $2x_t$ in an aquifer of thickness h and hydraulic conductivity K . Assuming that a drawdown of ΔP in the trench influences the aquifer to a distance a gives [Harr, 1962]

$$Q_{s1} = \frac{2Kx_t}{a} [2h\Delta P - \Delta P^2] \quad (1)$$

In contrast, considering the problem in plan view so that effects beyond the ends of the trench are included gives [Muskat, 1937]

$$Q_{s2} = \frac{2\pi\Delta PKh}{\ln\left(\frac{a+b}{x_t}\right)} \quad (2)$$

assuming that the drawdown is uniform along the entire trench and that the trench influences an elliptical area of minor axis a and major axis b (major axis is parallel to the trench). Assuming that $\Delta P/h = 0.1$ and $b = a + x_t$, the ratio of flow rates Q_{s1}/Q_{s2} for various values of a/x_t and b/x_t are given in Table 1.

Based on those parameters, the analyses yield similar results if the trench influences a distance that is short relative to its length ($a \ll x_t$). As the distance of influence increases to the same order as the length of the trench, however, the analysis that neglects effects of the ends of the trench markedly underestimates the discharge. Accordingly, the assumption of infinite length might provide a result of reasonable accuracy at early times when the trench affects a short distance into the adjacent aquifer. The accuracy of those results will diminish with increasing time as the trench affects increasingly large distances, and they will be poorest when conditions have reached steady state. This points out a dichotomy between the assumptions of steady state and infinite length (two-dimensional section view) analyses for interceptor trenches.

Analyses that consider a trench of finite length [e.g., Strack, 1989; Muskat, 1937] should provide a more accurate prediction of long-time behavior than analyses that assume infinite length. The available analyses that assume finite length are limited to steady state conditions, however, and are unable to show transient effects. As a result, there are no guidelines as to how long it takes before an interceptor trench will reach steady conditions. Moreover, transient conditions may occur for a long period, so the use of steady state analyses could predict values of drawdown and contaminant travel that exceed field values. Koch [1986] used a

Table 1. Comparison of Discharges From Steady State Analyses of Trenches

a/x_t	b/x_t	Q_{s1}/Q_{s2}
0.1	1.1	1.10
0.2	1.2	1.02
0.5	1.5	0.84
1.0	2.0	0.66
2.0	3.0	0.49
5.0	6.0	0.29
10.0	11.0	0.18

series of steady state solutions to approximate a transient response to a trench, but a fully transient analysis of an interceptor trench apparently has yet to be published.

This paper uses the method of instantaneous source functions to derive transient analytical solutions to the performance of a variety of two- and three-dimensional problems related to interceptor trenches. Here performance is taken as the drawdown as a function of time in the vicinity of a trench that is pumped at constant rate or the discharge from a trench that is held at constant drawdown. During remediation, performance also includes the path taken by contaminants (the particle path) and the duration of travel from a given point to the trench (the arrival time).

Method of Analysis

The performance of a single, straight trench during the recovery of groundwater will be evaluated by deriving analytical solutions to the problem of a rectangular sink in an idealized aquifer. We will assume that the head in the aquifer, p , satisfies the Poisson equation

$$\frac{\partial^2 p}{\partial x^2} + \frac{\partial^2 p}{\partial y^2} + \frac{\partial^2 p}{\partial z^2} = S/Kh \frac{\partial p}{\partial t} \quad (3)$$

where K is hydraulic conductivity, and S is a storage term (i.e., the storage coefficient of a confined aquifer or the effective porosity of an unconfined aquifer).

Instantaneous Source Functions

A solution to (3) for appropriate boundary conditions will be obtained using the method of instantaneous source functions [Gringarten and Ramey, 1973]. In general, this method uses superposition of a basic solution to build analytical expressions to problems of practical interest. The basic solution is for a point sink that is active during an instantaneous pulse of a given strength. That solution is integrated along x to obtain the instantaneous source function for a line and then integrated again along z to obtain the instantaneous source function for a plane. The integration procedure tacitly assumes that the strength, or flux, is uniformly distributed over the surface of the trench. No flow or constant head far-field boundaries are readily included by reflection. Accordingly, the geometric aspects of sources, sinks, and far-field boundaries are all included in the source function. To obtain a useful solution, the source function is integrated with time. The result will apply to cases of constant discharge if the strength of the source is independent of time, or it will apply to transient discharge if the strength varies with time.

This procedure gives the drawdown ΔP as a function of time as

$$\Delta P(x, y, z, t) = (p_i - p) = h/S$$

$$\cdot \int_0^t q(\tau) s(x, y, z, t - \tau) d\tau \quad (4)$$

where $s(x, y, z, t)$ is the source function, q is the source strength (i.e., discharge per unit length, area, or volume of the source), p_i is initial head, and p is the head at (x, y, z, t) .

Assumptions

We will assume that the trench penetrates an isotropic confined aquifer of thickness h and infinite lateral extent. The results will also be an approximation of conditions within an unconfined aquifer if the drawdown in the trench is small compared to the saturated thickness of the aquifer [Bear, 1979]. The hydraulic conductivity of the aquifer will be uniform and isotropic, although modifications to consider anisotropic conditions are straightforward [Gringarten and Ramey, 1973].

The interceptor trench itself will be idealized as a planar sink centered on the origin with half-length x_t , depth d , and negligible width (Figure 1). The flux will be assumed to be uniformly distributed over the surface of the trench, which tacitly requires that the head increases slightly from the midpoint to the end of the trench and implies that the trench is filled with a material of high, but finite, conductivity. This is a reasonable approximation to many field designs. It will be assumed that there is negligible drawdown an infinite distance from the trench, which ignores the effects of far-field boundaries. So, in general,

$$\frac{\partial p}{\partial y} = -q/K \quad -x_t \leq x \leq x_t, y = 0, h - d \leq z \leq h \quad (5a)$$

$$\Delta P = 0 \quad x \rightarrow \infty \text{ or } y \rightarrow \infty \quad (5b)$$

where q is uniform over the surface given above.

Special arguments will be presented to extend the uniform flux condition to predict the drawdown within a trench filled with material of infinite conductivity. In this case, there is no head variation along the trench, a situation that may approximate a large contrast between the conductivity of the aquifer and the conductivity of the trench. The uniform flux

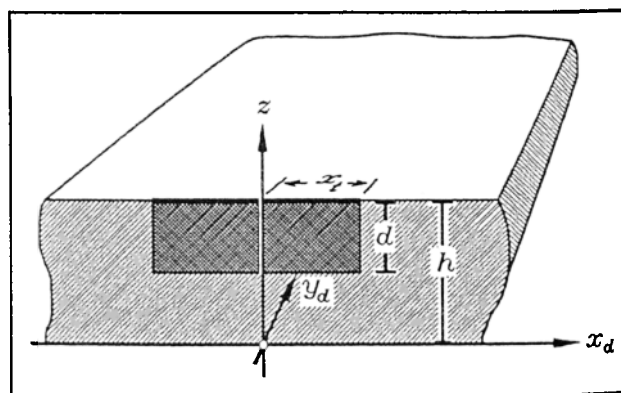


Figure 1. Configuration of interceptor trench used in the analyses.

and infinite conductivity cases are presented to highlight the effects of different conditions within the trench, although it should be pointed out that the hydraulics within the trench itself are not explicitly considered here. We simply assume that fluid arriving at the trench face is instantly removed from the system.

Two pumping scenarios will be considered. In one, the discharge from the trench is constant and we seek the drawdown at the origin, whereas in the other case the drawdown at the origin is constant and we seek the discharge from the trench. The assumption of flux uniformly distributed over the trench is maintained in both cases; however, the flux is constant in the former case, and it varies with time to maintain constant drawdown in the latter case. Accordingly, q is constant in the case of uniform flux, whereas $q = f(t)$, such that $\Delta P = \text{const}$ at $x = 0, y = 0, z = 0$ for the case of constant drawdown.

Drawdown During Constant Discharge

In this section, drawdown is given for a trench of finite length that partially penetrates an aquifer, and then that three-dimensional case is reduced to a two-dimensional problem by allowing the trench to completely penetrate the aquifer.

Drawdown Due to a Partially Penetrating Trench

The source function for a trench in an aquifer of thickness h (Figure 1) is given by the intersection of three source functions for (1) an infinite planar source in the x - z plane, (2) an infinite slab source in the y - z plane, and (3) an infinite slab source in the x - y plane that is confined by impermeable boundaries at $z = 0$ and $z = h$. The functions are given as I, II and X by *Gringarten and Ramey* [1973, Tables 1 and 2].

Using the dimensionless quantities

$$t_d = 4tKh/Sx_t^2 \quad (6a)$$

$$P_d = 4\pi hK\Delta P/Q \quad (6b)$$

$$x_d = x/x_t \quad (6c)$$

$$y_d = y/x_t \quad (6d)$$

$$z_d = z/x_t \quad (6e)$$

The source function for the trench is

$$\begin{aligned} s_3(x_d, y_d, z_d, t_d) = & 1/[x_t 2(\pi t_d)^{1/2}] \exp(-y_d^2/t_d) \\ & \cdot \{\text{erf}[(1+x_d)/t_d^{1/2}] + \text{erf}[(1-x_d)/t_d^{1/2}]\} \\ & \cdot \frac{d}{h} \left\{ 1 + \frac{4h}{\pi d} \sum_{n=1}^{\infty} \frac{1}{n} \exp\left(-\frac{n^2 \pi^2 t_d x_t^2}{4h^2}\right) \right. \\ & \cdot \sin \frac{n\pi d}{2h} \cos \frac{n\pi(h-d/2)}{h} \cos \frac{n\pi z_d x_t}{h} \end{aligned} \quad (7)$$

The dimensionless drawdown resulting from a constant rate of withdrawal (pumping) from the trench is obtained by first changing the variable of integration in (4):

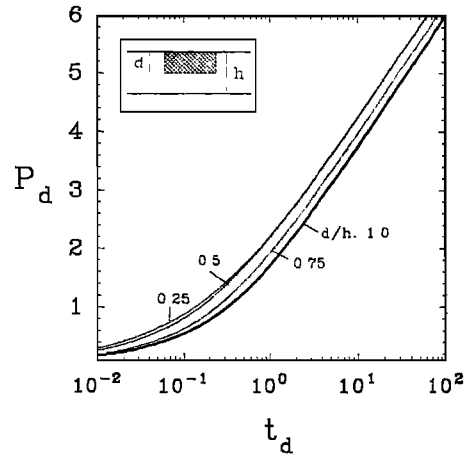


Figure 2. Drawdown at the origin as a function of time due to a partially penetrating trench of various depths.

$$\Delta P(x, y, z, t) = x_t^2/4K \int_0^{t_d} q(\tau) s(x, y, z, t - \tau) d\tau \quad (8)$$

The total discharge is

$$Q(t) = q(t)2x_t d \quad (9)$$

where $x_t 2d$ is the area of the trench. Substituting (8) into (9) and introducing the dimensionless form of the source function

$$s_{d3}(x_d, y_d, z_d, t_d) = x_t s_3(x, y, z, t) \quad (10)$$

we get using (6)

$$P_d(x_d, y_d, z_d, t_d) = \frac{h\pi}{2d} \int_0^{t_d} s_{d3}(x_d, y_d, t_d - \tau) d\tau \quad (11)$$

Dimensionless drawdown was evaluated using (7) and (11), assuming $x_t = h = 1.0$, and examining various ratios of d/h (Figure 2). In general, the drawdown at any time increases as the depth of the trench diminishes from a planar sink that fully penetrates the aquifer at $d/h = 1.0$ to a line sink across the top of the aquifer at $d/h = 0$. Interestingly, however, the form of the drawdown curve at moderate to late times ($t_d > 1.0$) is independent of the ratio d/h ; the drawdown approaches a logarithmic function of time. The drawdown from a vertical well also approximates a logarithmic function of time, indicating that the behavior of an interceptor trench, regardless of its depth relative to the thickness of the aquifer, will resemble a large well at relatively long times.

Drawdown in Two Dimensions, Plan View

In the analyses that follow, we will address the two-dimensional problem of a trench that fully penetrates an isotropic aquifer. This simplification allows us to eliminate the infinite series in (7) because $d = h$. The dimensionless source function reduces to

$$\begin{aligned} s_d(x_d, y_d, t_d) = & 1/[2(\pi t_d)^{1/2}] \exp(-y_d^2/t_d) \\ & \cdot \{\text{erf}[(1+x_d)/t_d^{1/2}] + \text{erf}[(1-x_d)/t_d^{1/2}]\} \end{aligned} \quad (12)$$

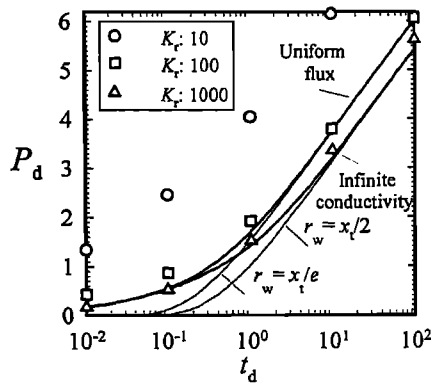


Figure 3. Drawdown at the origin as a function of time due to a trench of uniform flux and one of infinite hydraulic conductivity (thick lines). Drawdown due to wells whose diameters are scaled to the length of the trench (thin lines). Drawdown from trenches of finite conductivity (K_r , hydraulic conductivity of trench/hydraulic conductivity of formation) calculated numerically (symbols).

and drawdown anywhere in the system is given by substituting (12) into (11) and integrating. At the midpoint of the trench the drawdown reduces to (Figure 3)

$$P_d(0, 0, t_d) = (\pi t_d)^{1/2} \operatorname{erf}(1/t_d^{1/2}) + E_1(1/t_d) \quad (13)$$

where E_1 is the exponential integral [Abramowitz and Stegun, 1964]

$$E_1(x) = \int_x^\infty \frac{e^{-u}}{u} du \quad (14)$$

Gringarten and Ramey [1974] noticed that by setting $x_d = 0.732$ and $y_d = 0$ in (11) and (12) they obtained the drawdown for a planar sink of infinite conductivity at early and late times, and they argue that the drawdown is thus obtained for all times. Using the assumptions of Gringarten and Ramey [1974], the dimensionless drawdown for a trench of infinite conductivity is given by

$$P_d(t_d) = 0.5(\pi t_d)^{1/2} [\operatorname{erf}(0.268/t_d^{1/2}) + \operatorname{erf}(1.732/t_d^{1/2})] + 0.134E_1(0.072/t_d) + 0.866E_1(3.0/t_d) \quad (15)$$

This result produces a drawdown that is similar to the uniform conductivity case at early time, but is parallel to and approximately 0.6 less than the uniform flux case at late time (Figure 3).

The essence of the behavior of a trench with time is expressed in the form of (13). At early times the second term in (13) vanishes, and the error function term approaches unity, so that P_d is proportional to the square root of time. For example, P_d is within 1.0% of $(\pi t_d)^{1/2}$ when $t_d < 0.25$. Drawdown due to a planar source of infinite extent is proportional to the square root of time, indicating that the effects of the end of the trench are negligible when $t_d < 0.25$.

At long times the first term becomes constant, and the second term, E_1 , varies with time. This indicates that at long times the drawdown response of a trench resembles the exponential integral solution of Theis [1935] for a vertical well. Dimensionless drawdown when $t_d > 25$ for an infinite

conductivity trench is within 1% of $E_1(1/t_d)$, where instead of x_t in t_d we use a well of equivalent radius $r_e = x_t/2$ (Figure 3). Muskat [1937] and Prats [1961] both showed that the same equivalency between well radius and trench length occurs at steady state. Drawdown due to the uniform flux trench approaches that of a well whose radius is slightly smaller, $r_w = x_t/e$, where e is the base of the natural logarithm. The solutions are within 1% when $t_d > 5$ (Figure 3). The well of equivalent radius to a uniform flux trench at steady state is given in the appendix.

The analytical solutions were compared to numerical analyses that consider a trench as a band where the hydraulic conductivity is greater than the surrounding aquifer (Figure 3). The band had an aspect ratio of 40:1, and there was a well at the midpoint. The results indicate that the uniform flux solution resembles a trench where the ratio of hydraulic conductivity in the trench to that of the formation, K_r , is approximately 100. The infinite conductivity solution resembles the numerical results for $K_r \geq 1000$.

Discharge During Constant Drawdown

In some field applications the drawdown at the pump in a trench is held constant using a level-activated switch or similar device, and we wish to determine the discharge of the pump as a function of time. To do so, we define the dimensionless discharge as

$$Q_d(t) = \frac{Q(t)}{\Delta P_w 8Kh} \quad (16)$$

with ΔP_w the drawdown at the pump, and it follows that

$$\Delta P(x_d, y_d, t_d)/\Delta P_w = \int_0^{t_d} Q_d(\tau) s_d(x_d, y_d, t_d - \tau) d\tau \quad (17)$$

The dimensionless discharge is obtained by setting $x_d = y_d = 0$ so that

$$1 = \int_0^{t_d} Q_d(\tau) s_d(0, 0, t_d - \tau) d\tau \quad (18)$$

One method of approximating $Q_d(t)$ is to recognize that Q_d approaches $1/P_d$ as the system approaches steady state [Murdoch and Franco, 1994]. Introducing $g(t_d)$ as a correction term that accounts for the early time differences between a sink held at constant drawdown and one maintained at constant discharge gives

$$Q_d(t_d) = 1/P_d(t_d) + g(t_d) \quad (19)$$

We showed previously that early drawdown was proportional to the square root of time, so we will assume that $g(t_d)$ has a power law form

$$g(t_d) = a_1 t_d^{a_2} + a_3 t_d^{a_4} \quad (20a)$$

Constants that best satisfy (18) using (19) and (20a) were estimated using a Levenberg-Marquardt algorithm [Press et al., 1989]. Those coefficients are

$$a_1 = -0.414229 \quad (20b)$$

$$a_2 = -0.462656 \quad (20c)$$

$$a_3 = 0.243483 \quad (20d)$$

$$a_4 = -0.385682 \quad (20e)$$

which gives the dimensionless discharge in Figure 4. Draw-down anywhere in the aquifer is determined by substituting (19) into (17).

Dimensionless discharge from (19) and (20) was used in (18) to check how closely the condition of constant draw-down was satisfied. The average residual error is 0.0106 over the range $10^{-4} < t_d < 10^{11}$. The maximum predicted drawdown was 1.0365, whereas the minimum was 0.9582, and both those values occur in the period $0.25 < t_d < 25$. At early and late times the results are particularly accurate, with drawdown varying by less than 1% from 1.0 (Figure 4).

In some cases, such as when hand calculations are necessary, it is convenient to use a simpler function for $g(t_d)$. For example, requiring $g(t_d)$ to be proportional to the inverse of square root of time yields $g(t_d) = -0.2/t_d^{1/2}$, which has an average residual error of 0.0225, and maximum and minimum drawdowns that are 1.0159 and 0.8763, respectively. Although that expression is slightly less accurate than the one given in (20), most of the inaccuracy occurs in the transition period, and it predicts drawdowns that are within 1% of 1.0 during early and late times. The procedures described above can be used with other functions for $g(t_d)$ if greater accuracy is required.

It should be pointed out that the analysis used above compares the expected drawdown of 1.0 to the drawdown obtained by integrating the estimated $Q_d(t_d)$ through time in (18). The integration could accumulate or cancel any differences between the estimated and the exact Q_d . As a result, the differences between the estimated and exact Q_d are not necessarily the same as the differences between the estimated and exact drawdown that results from the Q_d .

Flow in the Vicinity of a Trench

Unlike radial flow to a well the pattern of flow to a trench changes with time, and the changing pattern is closely tied to trench performance. The developing flow pattern goes through three distinct periods: a linear period at early time, a transition period at intermediate time, and radial period at late time. *Cinco-Ley and Samaniego-V. [1981]* recognized

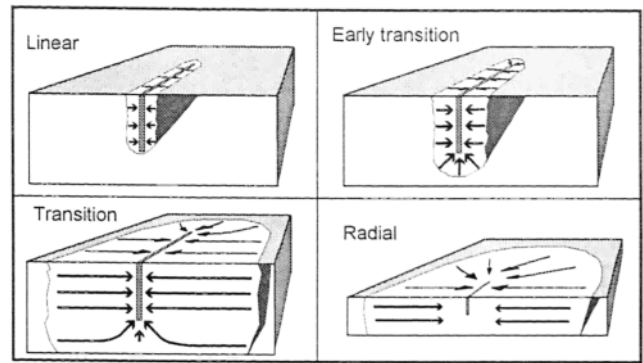


Figure 5. Conceptual model of flow to a trench during the linear, transition, and radial periods.

similar periods during transient flow to a vertical fracture. The period of linear flow involves fluid adjacent to the trench that follows straight paths nearly perpendicular to the plane of the trench (Figure 5). In contrast, the period of radial flow involves fluid that converges from large distances toward the trench. Details of the flow pattern during the transition period, as the flow changes from linear to radial, depend on the details of the trench geometry. For example, if the trench partially penetrates the aquifer, the linear period is followed by a period when the flow is linear adjacent to the trench, radial and converging upward beneath the bottom of the trench, or radial and converging horizontally toward the ends of the trench (Figure 5). With increasing time the lower radial zone expands downward and affects the bottom of the aquifer, and the area influenced by the trench increases (Figure 5). Eventually, the area of horizontal radial flow grows until it dominates and the period of radial flow begins. The transition period is somewhat simpler when the trench fully penetrates the aquifer because the early period of upward flow toward the bottom of the trench is absent.

The results presented earlier indicated that at early times ($t_d < 0.25$) the drawdown induced by a trench resembles that of planar sink of infinite extent, whereas at late times ($t_d > 25$) it resembles a large-diameter well. Accordingly, we will use the analysis of drawdown to define the durations of the flow periods.

Linear flow

$$t_d < 0.25 \quad (21a)$$

Transition

$$0.25 < t_d < 25 \quad (21b)$$

Radial flow

$$t_d > 25 \quad (21c)$$

where as before $t_d = 4tKh/Sx_i^2$.

The duration of the different flow periods in real time t can vary widely depending, in particular, on the hydraulic conductivity of the aquifer and length of the trench. Consider, for example, a trench 200 m long ($x_i = 100$ m), in a layer of silty sand that is 5 m thick, with $K = 10^{-4}$ cm/s and $S = 0.1$. The linear flow period would last approximately 0.4 years and it would be 40 years before the radial flow period began. However, if the trench were half that long ($x_i = 50$

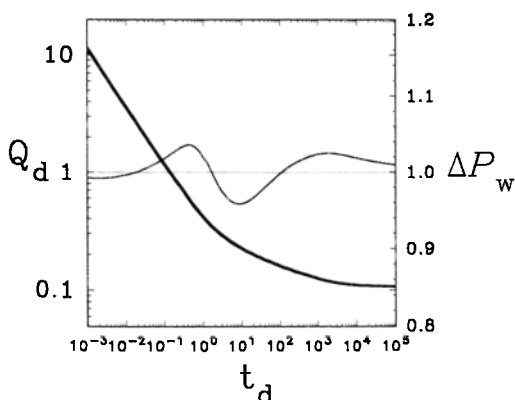


Figure 4. Dimensionless discharge as a function of time (thick line). Dimensionless drawdown at the midpoint of the trench calculated using Q_d (thin line).

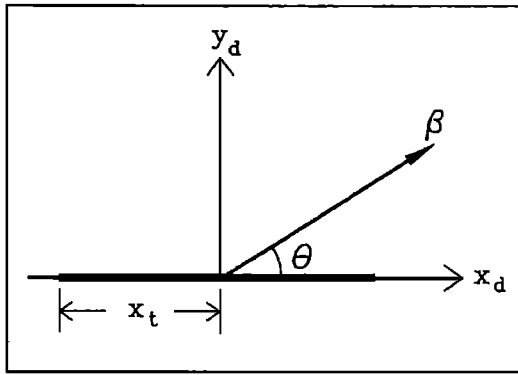


Figure 6. Configuration of the regional gradient.

m) and it was installed in a clean sand ($K = 10^{-2}$ cm/s), the linear flow period would last 8.7 hours and the radial flow period would begin after 36 days.

Particle Tracking

The paths of imaginary “particles” of water are commonly tracked through an aquifer to simulate advective transport of contaminants. The analyses presented here lend themselves to particle tracking by providing analytical expressions for particle velocity, which can be integrated to determine displacements. From Darcy’s law the velocity in the x direction of a particle in the flow field is

$$v_x = dx/dt = -[K/Rn]\{-\alpha \cos \theta + [Q/4\pi Khx_t]dP_d/dx_d\} \quad (22)$$

where n is porosity, R is a retardation factor, α is the magnitude of a regional head gradient, and θ is the angle between the regional gradient vector and the x axis (Figure 6). The dimensionless velocity is

$$v_{xd} = dx_d/dt_d = [Sx_t/4Kh] dx/dt = \beta \cos \theta - \Psi dP_d/dx_d \quad (23a)$$

with

$$\beta = x_t \alpha S/4hRn \quad (23b)$$

$$\Psi = QS/16Kh^2\pi Rn \quad (23c)$$

The quantities Ψ , β , and θ are dimensionless parameters that define stresses on the aquifer. The term Ψ is the dimensionless strength of the trench (discharge scaled to the capacity of the aquifer to provide water), β is the dimensionless strength of the regional gradient, and θ is the orientation of the regional gradient. The dimensionless gradient dP_d/dx_d can be obtained by reversing the order of integration and differentiation so that

$$dP_d/dx_d = \frac{\pi}{2} \int_0^{t_d} s'_{dy} dt \quad (24a)$$

where

$$s'_{dx} = 1/(\pi t_d) \exp(-y_d^2/t_d) \cdot \{\exp[-(1+x_d)^2/t_d] - \exp[-(1-x_d)^2/t_d]\} \quad (24b)$$

and similarly

$$dP_d/dy_d = \frac{\pi}{2} \int_0^{t_d} s'_{dy} dt \quad (25a)$$

with

$$s'_{dy} = -y_d/(\pi^{1/2} t_d^{3/2}) \exp(-y_d^2/t_d) \cdot \{\text{erf}[(1+x_d)/t_d^{1/2}] + \text{erf}[-(1-x_d)/t_d^{1/2}]\} \quad (25b)$$

Trajectories of particles were determined by integrating dimensionless velocity using a fourth-order Runge-Kutta algorithm. Particle tracking was facilitated by decreasing the time step used in the Runge-Kutta algorithm as the particle approached the trench. This was done because large time steps were used early in the tracking history to rapidly step through time when the effects of the trench were negligible. The size of the time step was diminished as the particle began to move, however, to limit the size of the displacement; constant time steps will increase the size of the displacement and compromise the accuracy of the procedure. The duration of each time step was selected by dividing the desired displacement by an estimate of the velocity at the particle location.

Fields of particles were tracked from their initial positions until they arrived at the trench, and this time was taken as the arrival time, t_{da} . Arrival times were determined for various Ψ along two radial paths that intersect the origin; one path was parallel, whereas the other was normal to the trench (Figure 7). A regional gradient was assumed to be absent, and several different source strengths Ψ were evaluated. The path parallel to the trench yields arrival times that are earlier than the path normal to the trench. At large distances from the trench the two plots become nearly parallel, which is consistent with transport by radial flow. However, t_{da} cannot be determined by analogy to a single equivalent well, because the radial path parallel to the trench will always be shorter and give t_{da} that is less than the path parallel to the trench.

Arrival time depends strongly on Ψ , with an increase in Ψ decreasing the arrival time from any given point. For a

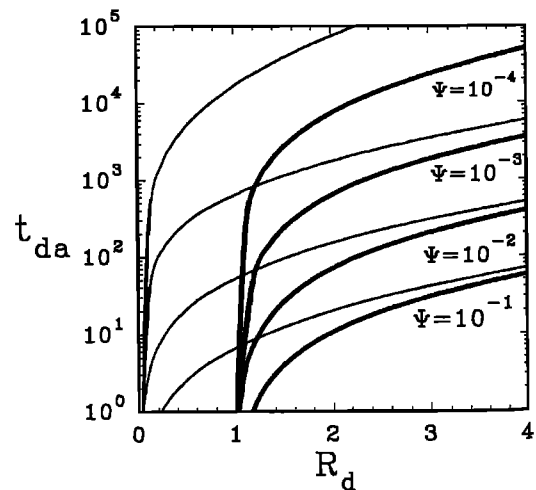


Figure 7. Dimensional arrival time as a function of radial distance parallel to the trench (thick line) and perpendicular to the trench (thin line) for various values of Ψ . Regional flow is absent. R_d equals radial distance divided by x_t .

particular set of aquifer conditions, however, increasing Ψ can only be accomplished by increasing discharge. This will increase drawdown, which will be limited during field applications by the depth of the trench. As a result, drawdown will place a practical upper limit on the value of Ψ that is not explicitly considered in the analysis.

Geometrical effects are particularly strong in the vicinity of the trench, according to Figure 8, where the radial coordinate system of Figure 7 is used with different values of angular position, γ . The arrival time depends strongly on the starting location, particularly where $R_d < 1.25$ and the geometry of the trench controls the details of the flow field.

The effects of regional flow are shown by examining values of $\beta = 0, -0.003, -0.01$, and -0.03 and values of $\theta = 0, \pi/4$, and $\pi/2$, with $\Psi = 0.01$ in all cases. The general problem of a trench with regional flow lacks symmetry, so the fields of arrival times were contoured to produce maps in Figure 9.

Introducing regional flow causes some particles to be swept away and never reach the trench. The capture zones form U-shaped zones, with the principal axis in the direction of regional flow. The shape of the capture zone changes with increasing β by shortening in the downgradient and lengthening in the upgradient dimension. Shortening of the capture zone also occurs normal to the regional flow, but to a lesser extent than shortening downgradient. Keep in mind, however, that the contours of equal t_{da} can change shape but they cannot change area as a result of changing the direction of a regional flow that is uniform and constant. This occurs because contours of equal t_{da} must circumscribe the same volume of water in a system that is being operated at constant discharge [Keely, 1984].

Application of the maps of arrival times is illustrated by assuming a hypothetical situation where a trench is installed in an aquifer with the following parameters: $K = 10^{-4}$ m/s; $S = 0.1$; $n = 0.1$; $h = 5$ m, and $x_t = 50$ m. For $\Psi = 0.01$, it follows that $Q = 0.16\pi nKh^2/S = 0.07539$ m³/min (20 gallons/min). Assuming the regional gradient is 0.004, then $\beta = -0.01$ and the map at the bottom of the middle column in Figure 9 represents the site conditions. The arrival times for specific points can be estimated by obtaining t_{da} from Figure

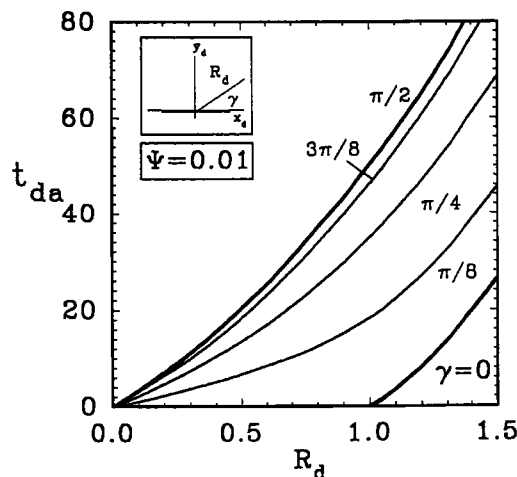


Figure 8. Detail of arrival time as a function of location in the vicinity of a trench. Regional flow is absent. R_d equals radial distance divided by x_t .

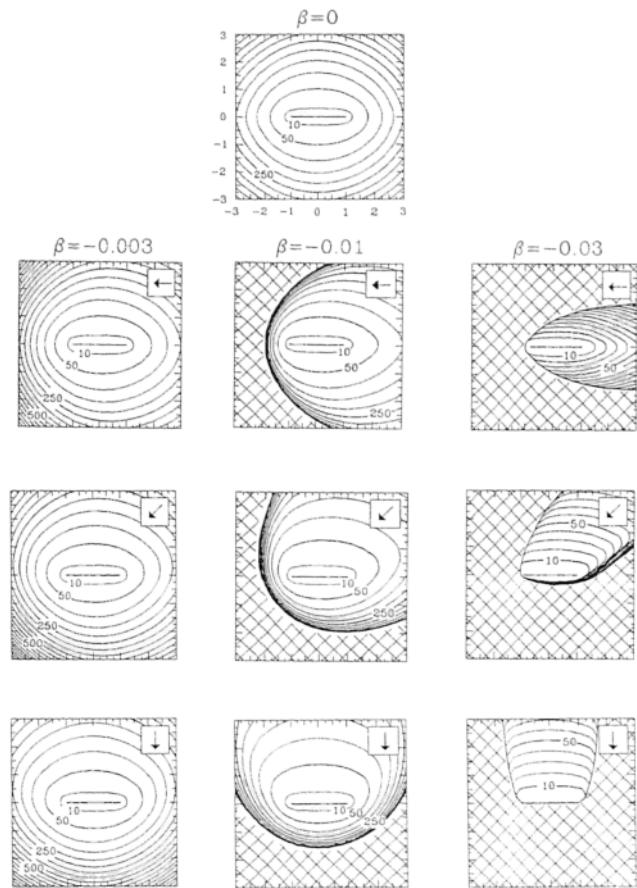


Figure 9. Dimensionless arrival times as a function of location for various strengths and directions of regional flow. $\Psi = 0.01$. Particles are swept away by regional flow in cross-hatched areas.

9. For example, a point $x = 50$ m, $y = 50$, where y is up the regional gradient ($x_d = 1.0$, $y_d = 1.0$), yields $t_{da} = 38$, which is 55 days in real time. The effect of the regional gradient is illustrated by taking a point of equal distance down the regional gradient ($x_d = 1.0$, $y_d = -1.0$), where $t_{da} = 164$, or 237 days. Alternatively, the area affected after 6 months is circumscribed by the contour of $t_{da} = 126$.

The maps in Figure 9 can be applied to most aquifer settings where the limitations set by assuming confined conditions are valid. The effect of regional gradient is negligible and probably can be ignored for $|\beta| < 0.001$. At large values of regional gradient ($|\beta| > 0.05$), the trench operating at a strength of $\Psi = 0.01$ is unable to generate a strong enough sink to effectively capture particles. The maps given in Figure 7 are limited to $\Psi = 0.01$, so other strengths will require calculating other maps.

It is noteworthy that interceptor trenches oriented at acute angles to the regional flow may be valuable for contaminant recovery. An interceptor trench oriented parallel to the regional gradient is an effective sink for areas upgradient of the trench. Accordingly, in areas where access or other factors prevent installing a trench normal to the regional flow, it may be possible to achieve the desired performance by installing the trench oblique to the flow. Where the regional gradient is inclined $\theta = 45^\circ$ to the axis of the trench,

the arrival times in the region upgradient of the trench are similar to those where $\beta = 90^\circ$ (Figure 9).

Conclusions

The flow to a straight trench in an idealized confined aquifer in the absence of regional gradient occurs in three distinct patterns: linear, transition, and radial, which evolve from one to another with increasing duration of pumping. During early times, termed the linear flow period when $t_d < 0.25$, the flow occurs in a narrow zone adjacent to the trench. Flow trajectories are nearly straight and parallel, resembling one-dimensional flow to a planar sink of infinite dimensions. Solutions that depict a trench in cross section by assuming that it is infinitely long are valid during the linear flow period.

A period of radial flow occurs at long times, $t_d > 25$, when flow converges from great distances and the trench behaves like a well whose diameter is of the order of the half-length of the trench. Drawdown at constant discharge and discharge at constant drawdown for a trench during the radial flow period can be approximated by wells of large diameter.

The transition period occurs when $0.25 < t_d < 25$ as the pattern of flow adjusts from linear to radial. There is no simple analogy to the flow pattern during the transition period.

The duration of the flow periods in real time depends on the trench length and aquifer properties. Consequently, the linear flow period can be finished in a few minutes or it can last many years. The changing pattern of flow is critical to predicting the advective transport of contaminants, where contaminant displacement is determined by integrating a velocity field that varies with both time and space. Regional gradients, recharge, or lateral aquifer boundaries will distort the pattern of flow, but the effect of those conditions was ignored when identifying the flow periods.

The method of instantaneous source functions [Gringarten and Ramey, 1973] was used to provide the drawdown at constant discharge and discharge at constant drawdown as functions of time. Those results can be used with aquifer properties to estimate the transient performance of an interceptor trench. Conversely, the analytical solutions can be used with pumping records to estimate aquifer properties, just as related solutions are used to interpret pumping tests at wells.

The analyses presented here are limited to a single straight trench, but many field applications include a network of trenches. Multiple, straight segments of trenches can be simulated by superposing the solution for a single trench of constant discharge presented here. Direct superposition of the solution for a trench at constant drawdown, however, is inappropriate because it will violate the conditions of constant head. The discharge from multiple trench segments where one point is held at constant head can be evaluated, however, by superimposing the source functions for each segment to obtain a new s_d and then using a parameter estimation procedure with (18).

Appendix: The Steady State Well That Is Equivalent to a Uniform Flux Trench

A trench of length $2x_t$ that fully penetrates a confined aquifer of infinite extent will be analyzed as a vertical slab

(plane of finite width) sink of width $2w$. The total discharge Q results from flux that is evenly distributed over the surface of the trench. The total head at the midpoint of the trench is p_p . Integrating the solution for a vertical line sink [Muskat, 1937] from $-x_t$ to x_t gives

$$p = C_1 \left\{ y \left[\tan^{-1} \left(\frac{x + x_t}{y} \right) - \tan^{-1} \left(\frac{x - x_t}{y} \right) \right] + \frac{x + L}{2} \cdot \ln [(x + x_t)^2 + y^2] - \frac{x - x_t}{2} \ln [(x - x_t)^2 + y^2] \right\} + 2L(C_2 - C_1) \quad (A1)$$

where C_1 and C_2 are constants that are determined by assuming that the total discharge Q is

$$Q = -2Kh \int_{-x_t}^{x_t} q \, dx = -2Kh \int_{-x_t}^{x_t} \frac{dp}{dy} \Big|_{y=w/2} dx \quad (A2)$$

$$p_s = p_p \quad x = 0, y = w/2 \quad (A3)$$

Solving for the constants and assuming negligible width,

$$C_1 = \frac{Q}{4\pi x_t Kh} \quad w = 0 \quad (A4a)$$

$$C_2 = \frac{Q[1 - \ln(x_t)]}{4\pi Kh x_t} + \frac{p_p}{2x_t} \quad w = 0 \quad (A4b)$$

The dimensionless head for a vertical planar sink is

$$P_d = \frac{2\pi Kh(p_s - p_p)}{Q} = (y/2x_t) \cdot \left[\tan^{-1} \left(\frac{x + x_t}{y} \right) - \tan^{-1} \left(\frac{x - x_t}{y} \right) \right] + \frac{x + x_t}{4x_t} \ln [(x + x_t)^2 + y^2] - \frac{x - x_t}{4x_t} \ln [(x - x_t)^2 + y^2] - \ln(x_t) \quad (A5)$$

As x approaches infinity, the dimensionless head along $y = 0$ becomes

$$P_{d(x \rightarrow \infty)} = \{\ln(x) + [1 - \ln(x_t)]\} \quad (A6)$$

which is the same form as a vertical well [Theim, 1906] with an equivalent radius r_e given by

$$r_e = x_t/e \quad (A7)$$

Notation

- a minor axis of area influenced by a trench.
- b major axis of area influenced by a trench.
- d depth of a partially penetrating trench.
- h aquifer thickness.
- $g(t_d)$ function used to obtain Q_d .
- K hydraulic conductivity.
- n porosity.
- p head.

- p_i initial head.
 ΔP drawdown, equal to $p_i - p$.
 ΔP_w drawdown at the midpoint of the trench.
 P_d dimensionless drawdown.
 q source flux, discharge per unit area of trench.
 Q total discharge.
 Q_{s1} steady discharge from trench neglecting end effects.
 Q_{s2} steady discharge from trench including end effects.
 Q_d dimensionless discharge, equal to $Q/\Delta P_w 8Kh$.
 R retardation factor.
 R_w radius of influence of a well.
 R_d radial distance divided by x_f .
 R_e effective radius of influence of a trench.
 S storage coefficient.
 s source function.
 s_d dimensionless source function.
 t_d dimensionless time, equal to $4tKh/Sx_f^2$.
 t_{da} dimensionless time when particles arrive at trench.
 v_x fluid velocity.
 v_{xd} dimensionless velocity.
 x_t half-length of a trench.
 x, y, z spatial coordinates.
 x_d, y_d, z_d, R_d dimensionless coordinates scaled to x_t , equal to x/x_t , etc.
 α regional groundwater gradient.
 β strength of regional gradient, equal to $x_t \alpha S / 4hRn$.
 Ψ strength of trench, equal to $QS/16Kh^2 \pi Rn$.
 θ angle between trench and direction of regional gradient.

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